Student Performance Prediction using Machine Learning

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Project Overview

This project predicts student academic performance using Machine Learning techniques. The goal is to analyze student-related factors and generate predictions that can help identify performance trends and support data-driven educational decisions.

Objectives

- Predict student performance using historical data.
- Apply Machine Learning algorithms for analysis.
- Improve decision-making through data insights.

Technologies Used

Python, Pandas, NumPy, Scikit-learn, FastAPI, Next.js, Streamlit

Methodology

The dataset was preprocessed, cleaned, and analyzed. Relevant features were selected and a Machine Learning model was trained to predict student performance. The model was evaluated using standard performance metrics and integrated into a web-based application.

Results

The model successfully predicts student performance based on input parameters and demonstrates the practical use of Machine Learning in educational analytics.

Project Links

GitHub: https://github.com/2403a52007/Student-Performance-Ai
Live Demo: https://student-performance-ai-jvba3ljx7nbwrc4thrpkzg.streamlit.app/

Conclusion

This project highlights how Machine Learning can be used to predict student outcomes and support better educational planning. It demonstrates practical skills in data analysis, model development, and deployment.